Rakshit Ratan Pathak

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SUMMARY

Final-year B.Tech student (CGPA 8.3) who builds tested backend and systems software: a concurrency-safe PostgreSQL booking backend with 51 integration tests, a Java bug-repair agent, a crash-safe LSM-tree store and a C++17 compiler. Software engineering intern at aiDataWorks (LLM and RAG features on Google Cloud). Six patent applications, five published.

EDUCATION

Vellore Institute of Technology, Vellore | B.Tech, Computer Science and Business Systems | **CGPA 8.3** 2023–2027

Class XII (CBSE), Delhi Public School Ranchi: 87% | Class X (CBSE), DAV Jharsuguda: 99.8%

Relevant coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Computer Organization and Architecture.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++ (C++17/20), TypeScript, JavaScript, SQL
- **Backend and data:** Node.js, Express, Flask, FastAPI, REST APIs, PostgreSQL, MySQL, Redis, transactions and concurrency control, transactional outbox, role-based access
- **Systems:** storage engines (write-ahead log, LSM trees, compaction), compilers (parsing, ASTs, bytecode VMs), CUDA, sanitizers
- **Tools:** Git, Docker, GitHub Actions, CMake, pytest, Vitest, Linux, Google Cloud, Oracle Cloud (OCI), Vercel

PROJECTS

agentic-bug-repair: autonomous bug-fixing agent (Java 17, no dependencies) GitHub

- Takes a Java repo with failing tests, indexes it into a code graph (JDK compiler API), ranks likely bug locations, proposes small diffs and verifies each in a sandbox, then re-checks the winning diff on a clean checkout. GitHub webhook trigger and live dashboard.
- Fixed 11 of 17 seeded bugs without a model (11 of 12 single-edit bugs). About 5,700 lines of Java with 147 tests.

Healthcare Appointment and Follow-up Manager (TypeScript, Node.js, PostgreSQL, Redis) GitHub

- Built a REST backend with role-based access for patients, doctors and admins. Prevented double-booking with PostgreSQL exclusion constraints and 5-minute slot holds.
- Made notifications reliable with a transactional outbox, retries and a dead-letter queue, and added LLM visit summaries with a deterministic fallback. 51 integration tests run against a real Postgres.

mini-kv-store: crash-safe LSM-tree key-value store (Java) GitHub

- Write-ahead log with checksums, memtable, SSTables with sparse indexes and Bloom filters, k-way merge compaction and an atomic manifest. 39 tests, including crash injection at every flush and compaction step.

minic: a compiler in C++17 github.com/RakshitRPathak/minic

- 1,200+ line pipeline: lexer, recursive-descent parser, semantic analysis, optimizer, bytecode generator with backpatching and a stack VM. 43 tests, clean under AddressSanitizer and UBSan.

Also on GitHub: system-design-primitives (LRU/LFU caches, rate limiters, consistent hashing; 32 tests), dsa-patterns (52 patterns tested against brute force), VeriCloud (claim verification) and an inventory API with a hash-chained audit log.

EXPERIENCE

Software Engineering Intern, aiDataWorks (remote) May–Jul 2026

- Built and tested the LLM layer of a healthcare product: prompts, retrieval and context for a RAG pipeline on OpenAI APIs and enterprise Google Cloud.
- Wrote API- and database-level test cases for the LLM features and re-verified fixes with developers.
- Wrote data-governance and security policies for three client engagements, including mainframe data at a Tier 1 health insurer and a credential migration off Windows Server 2012.

Computer Science Intern, THDC India Limited (Khurja Super Thermal Power Project) May–Jun 2025

- Worked on SQL and SAP ERP data workflows in Python and Pandas, plus plant networking, SCADA/PLC systems and cybersecurity controls.

PATENTS, CERTIFICATIONS AND LEADERSHIP

- Six patent applications filed through VIT's IPR Cell, five published (opinion-lineage sentiment correction, IS trust-decay auditing, wildfire risk prediction, mine tremor detection, risk-constrained emergency control).
- Certifications: OCI Generative AI Professional, OCI AI Foundations, Google Cloud Core Infrastructure, Google AI Essentials.
- Secretary, IEEE Power & Energy Society student chapter, VIT. Karate black belt and assistant instructor.